

Exploratory Data Analysis Report for Global Electronics

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1. Introduction

Global Electronics is a leading retailer of consumer electronics with operations spanning multiple regions. To remain competitive and drive business growth, the company seeks to understand its customers, product performance, sales trends, and store operations better. This exploratory data analysis (EDA) provides a comprehensive review of key insights to enhance customer satisfaction, optimize inventory management, and improve sales forecasting.

2. Objectives

The primary objectives of this EDA are:

- **Customer Analysis:** Understanding customer demographics and purchasing behavior.
 - **Sales Analysis:** Analyzing sales performance by product, store, and currency impact.
 - **Product Analysis:** Evaluating product popularity and profitability.
 - **Store Analysis:** Assessing store performance based on sales, size, and location.
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3. Data Cleaning and Preparation

3.1. Data Sources

The data provided includes several datasets:

- **Customer Data:** Information on customer demographics.

- **Sales Data:** Sales transactions across multiple regions and currencies.
- **Product Data:** Details of products sold (e.g., category, price, cost).
- **Store Data:** Information on store size, location, and operational data.
- **Currency Exchange Rates:** Data for converting sales into a common currency.

3.2. Data Cleaning Process

- **Handling Missing Values:** Missing values were identified and handled appropriately using imputation or removal depending on the case.
- **Data Type Conversions:** Dates were converted to proper date formats, and numerical values were standardized. Categorical variables were transformed for easier analysis.
- **Merging Datasets:** The datasets were merged based on common keys such as customer ID, product ID, and store ID to facilitate analysis.

4. SQL Database Integration

The preprocessed data was loaded into an SQL database by creating relevant tables:

- **Customers** : Holds demographic details of customers.
- **Sales** : Contains sales transactions, linked with products and customers.
- **Products** : Stores product details, including category, price, and cost.
- **Stores** : Stores information on store locations and operations.
- **Currency_Exchange** : Holds exchange rates for converting foreign currency sales.

4.1. Example SQL Queries

Here are 10 key SQL queries developed to extract insights:

1. Total Sales Over Time:

```
SELECT YEAR(order_date), SUM(total_amount) AS Total_Sales
FROM Sales
```

```
GROUP BY YEAR(order_date)
ORDER BY YEAR(order_date);
```

Insight: Provides a high-level view of total sales by year.

2. Top-Selling Products by Revenue:

```
SELECT product_id, SUM(total_amount) AS Revenue
FROM Sales
GROUP BY product_id
ORDER BY Revenue DESC
LIMIT 10;
```

Insight: Highlights the products generating the highest revenue.

3. Customer Purchase Frequency:

```
SELECT customer_id, COUNT(order_id) AS Purchase_Frequency
FROM Sales
GROUP BY customer_id
ORDER BY Purchase_Frequency DESC;
```

Insight: Identifies customers who make frequent purchases.

4. Product Profitability (Margin Analysis):

```
SELECT product_id, SUM((unit_price - unit_cost) * quantity) AS Total_Profit
FROM Sales
JOIN Products ON Sales.product_id = Products.product_id
GROUP BY product_id
ORDER BY Total_Profit DESC;
```

Insight: Shows which products provide the highest profit margins.

5. Sales by Region/Store Location:

```
SELECT store_location, SUM(total_amount) AS Total_Sales
FROM Sales
JOIN Stores ON Sales.store_id = Stores.store_id
GROUP BY store_location
ORDER BY Total_Sales DESC;
```

Insight: Provides an overview of sales performance by store location.

6. Sales Seasonality:

```
SELECT MONTH(order_date), SUM(total_amount) AS Total_Sales
FROM Sales
GROUP BY MONTH(order_date)
ORDER BY MONTH(order_date);
```

Insight: Analyzes monthly sales trends for seasonality.

7. Average Order Value per Customer Segment (Age Group):

```
SELECT age_group, AVG(total_amount) AS Average_Order_Value
FROM Sales
JOIN Customers ON Sales.customer_id = Customers.customer_id
GROUP BY age_group;
```

Insight: Determines how different age groups contribute to average order value.

8. Currency Impact on Sales:

```
SELECT currency, SUM(total_amount * exchange_rate) AS Total_Sales_USD
FROM Sales
JOIN Currency_Exchange ON Sales.currency = Currency_Exchange.currency
GROUP BY currency;
```

Insight: Evaluates the impact of exchange rates on international sales.

9. Top Product Categories by Sales Volume:

```
SELECT category, SUM(quantity) AS Total_Quantity_Sold
FROM Sales
JOIN Products ON Sales.product_id = Products.product_id
GROUP BY category
ORDER BY Total_Quantity_Sold DESC;
```

Insight: Shows which product categories are selling the most.

10. Store Performance (Sales per Square Meter):

```
SELECT store_id, SUM(total_amount)/store_size AS Sales_Per
_Square_Meter
FROM Sales
JOIN Stores ON Sales.store_id = Stores.store_id
GROUP BY store_id
ORDER BY Sales_Per_Square_Meter DESC;
```

Insight: Identifies the most efficient stores based on sales per square meter.

5. Analysis Findings

5.1. Customer Analysis

- **Demographic Distribution:** The majority of Global Electronics customers are within the 25-44 age group, with a near-equal split between male and female customers. Geographically, most customers reside in urban areas, with North America contributing the largest customer base.
- **Purchase Patterns:** The average order value is highest among customers aged 35-44, indicating a more affluent demographic. Customers in urban areas show higher purchase frequencies, suggesting concentrated marketing efforts in these areas.
- **Customer Segmentation:** Three distinct customer segments were identified:

1. **Frequent Shoppers (High Purchase Frequency)**
2. **Big Spenders (High Average Order Value)**
3. **Occasional Shoppers (Low Purchase Frequency)**

5.2. Sales Analysis

- **Overall Sales Performance:** Sales show a significant upward trend over the last three years, with noticeable seasonal peaks during Q4 (holiday season).
- **Sales by Product:** Consumer electronics such as smartphones and laptops are the top-performing products, contributing to over 40% of total sales.
- **Sales by Store:** Stores in urban regions consistently outperform rural stores, with the highest sales observed in North America.
- **Currency Impact:** Currency exchange rates play a critical role in international sales, with significant fluctuations seen in regions using weaker currencies against the US dollar.

5.3. Product Analysis

- **Product Popularity:** Smartphones, laptops, and tablets are the most popular products. Home appliances and accessories are among the least popular.
- **Profitability Analysis:** The most profitable products include premium smartphones and high-end laptops, which offer better margins compared to budget-friendly options.
- **Category Analysis:** The "Electronics" category drives most of the sales, while smaller categories like "Accessories" contribute the least.

5.4. Store Analysis

- **Store Performance:** Stores in high-density urban areas outperform others in terms of total sales. Stores in regions with larger square footage tend to have lower efficiency compared to smaller stores in urban centers.
 - **Geographical Analysis:** North America and Europe lead in sales volume, while South America and Asia exhibit potential for growth but currently underperform.
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6. Visualizations

Interactive visualizations were created using Power BI/Tableau, offering a clear visual representation of the insights uncovered:

- **Sales Performance Over Time**
 - **Top Products by Revenue**
 - **Sales by Store Location**
 - **Customer Segmentation**
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7. Recommendations

Based on the findings from the analysis, the following recommendations are made to Global Electronics:

1. **Tailored Marketing Campaigns:** Focus marketing efforts on the 25-44 age group, especially targeting high spenders and frequent shoppers. Urban centers should receive more attention due to higher sales volumes.
 2. **Product Development:** Prioritize the development and promotion of high-margin products like premium smartphones and laptops, while reconsidering inventory for underperforming categories like accessories.
 3. **Inventory Optimization:** Leverage sales data to improve inventory management in underperforming regions and stores, ensuring stock availability for popular products.
 4. **Currency Strategy:** Consider adjusting international pricing strategies to account for fluctuations in exchange rates, especially in regions with weaker currencies.
 5. **Store Expansion:** Consider expanding store operations in regions with high growth potential, such as Asia and South America, while optimizing store layouts for better efficiency.
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8. Conclusion

The exploratory data analysis provided Global Electronics with key insights into customer demographics, sales trends, product performance, and store operations. By implementing data-driven recommendations, Global Electronics can increase customer satisfaction, optimize inventory management, improve sales forecasting, and enhance overall business growth.
